boot-flash

File: boot-flash.kicad_sch

core-boot-support

File: core-boot-support.kicad_sch

cpu-power

File: cpu-power.kicad_sch

ddr-bypass

File: ddr-bypass.kicad_sch

ddr3-memory

File: ddr3-memory.kicad_sch

display-power

File: display-power.kicad_sch

display-signals

File: display-signals.kicad_sch

ethernet-esd

File: ethernet-esd.kicad_sch

ethernet-magnetics

File: ethernet-magnetics.kicad_sch

ethernet-phy

File: ethernet-phy.kicad_sch

ethernet-support

File: ethernet-support.kicad_sch

pmic-bucks

File: pmic-bucks.kicad_sch

pmic-control

File: pmic-control.kicad_sch

pmic-ldos

File: pmic-ldos.kicad_sch

power-source-model

File: power-source-model.kicad_sch

soc-01

File: soc-01.kicad_sch

soc-02

File: soc-02.kicad_sch

soc-03

File: soc-03.kicad_sch

soc-04

File: soc-04.kicad_sch

soc-05

File: soc-05.kicad_sch

soc-06

File: soc-06.kicad_sch

soc-07

File: soc-07.kicad_sch

soc-core-bypass

File: soc-core-bypass.kicad_sch

soc-io-bypass

File: soc-io-bypass.kicad_sch

system-3v3

File: system-3v3.kicad_sch

test-access

File: test-access.kicad_sch

usb-analog-bypass

File: usb-analog-bypass.kicad_sch

usb-c-frontend

File: usb-c-frontend.kicad_sch

usb-input-protection

File: usb-input-protection.kicad_sch

usb-power-control

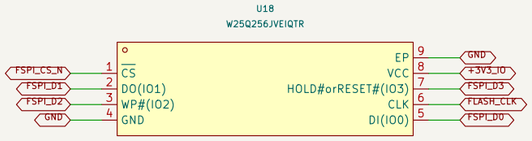
File: usb-power-control.kicad_sch

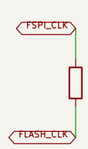
Not for fabrication

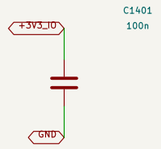
Sheet: /
File: phoenix-g1.kicad_sch

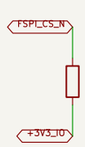
**Title: Phoenix G1 – integrated engineering draft**

| Size: A3 | Date: | **Rev: DRAFT** |
|---|---|---|
| KiCad E.D.A. 10.0.7 | | Id: 1/31 |



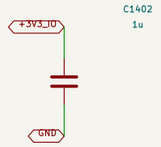



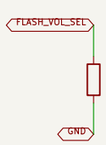










Candidate: W25Q256JVEIQ, 256Mbit = 32MiB. JLC C97522 inventory captured separately.

EP pad 9 grounded per Winbond technical FAQ; no exposed vias beneath its solder land.

32MiB requires a measured Linux/rootfs image budget. No A/B update capacity is claimed.

Boot ROM, loader, QE bit and 4-byte addressing compatibility need validation on hardware.

CS pull-up follows flash VCC. Clock resistor belongs at SoC; decoupling belongs at flash.

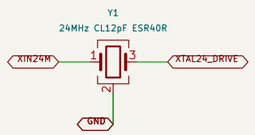
FLASH_VOL_SEL=0 selects 3.3V. Do not substitute a 1.8V flash without reworking the complete bank.

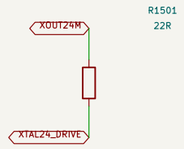
Not for fabrication

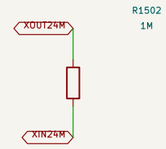
Sheet: /boot-flash/
File: boot-flash.kicad_sch

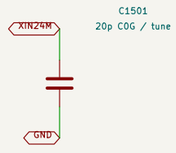
**Title: Phoenix G1 – 32MiB SPI NOR boot candidate**

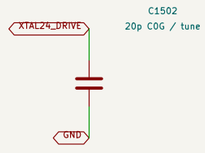
| | | |
|---|---|---|
| Size: A3 | Date: | **Rev: DRAFT** |
| KiCad E.D.A. 10.0.7 | | Id: 2/31 |

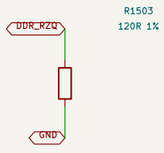


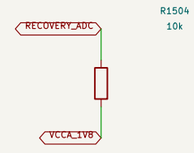



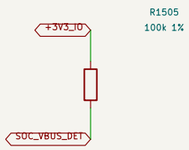



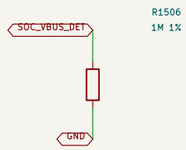



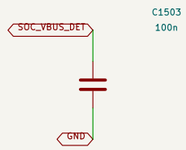




VBUS detect derives from USB-powered IO rail: 3.0V nominal; only valid for this single-source UFP topology.

JLC C2933041: 24MHz, CL12pF, ESR40R. 20pF assumes 2pF stray; frequency/drive require prototype tuning.

Recovery test point must ground RECOVERY_ADC during boot. UART is 3.3V logic, never RS232.

THERMAL_SHUTDOWN requires explicit reset wiring and polarity review before release.

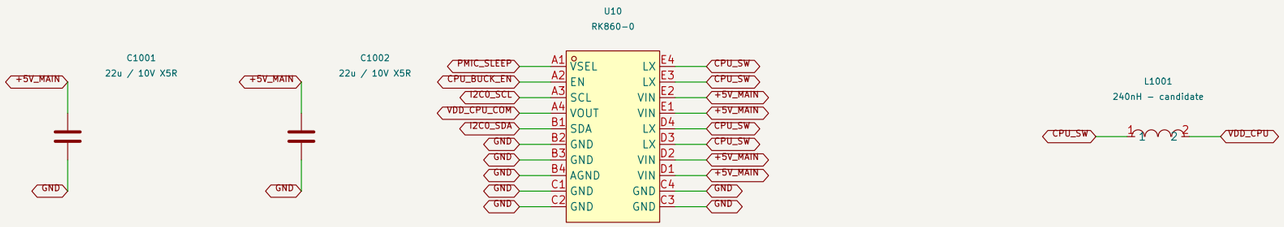
No external 32kHz crystal: PMIC internal RC selected; RTC accuracy is not specified.

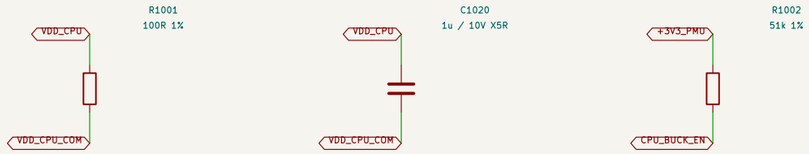
Not for fabrication

Sheet: /core-boot-support/
File: core-boot-support.kicad_sch

**Title: Phoenix G1 — oscillator, recovery and calibration**

| Size: A3 | Date: | **Rev: DRAFT** |
|---|---|---|
| KiCad E.D.A. 10.0.7 | | Id: 3/31 |




CPU POWER - DRAFT. Not a complete or release-qualified power system.

RK860-0: I2C address 0x40, DIE_ID=0x8, default 1.000 V, 12.5 mV step. No substitution with -2/-3.

VDD_CPU_COM is the RK3568 M15 remote-sense ball. Route separately from the high-current output path.

R1001 and C1020 bridge VDD_CPU to VDD_CPU_COM; C1020 is NOT a feedback-to-ground capacitor.

Input +5V_MAIN must be current-qualified and inrush-limited. +3V3_PMU enables CPU at the PMIC stage.

Add 4 x 22uF at the SoC CPU balls; total nominal CPU output capacitance becomes 176uF.

Effective capacitance must exceed the vendor requirement after tolerance, bias and temperature; MPNs pending.

Inductor candidate must pass Isat/Irms, ripple and thermal review; catalog current alone is insufficient.

I2C pull-ups and PMIC_SLEEP bias belong to the PMIC/SoC circuit; do not add duplicate pull-ups here.

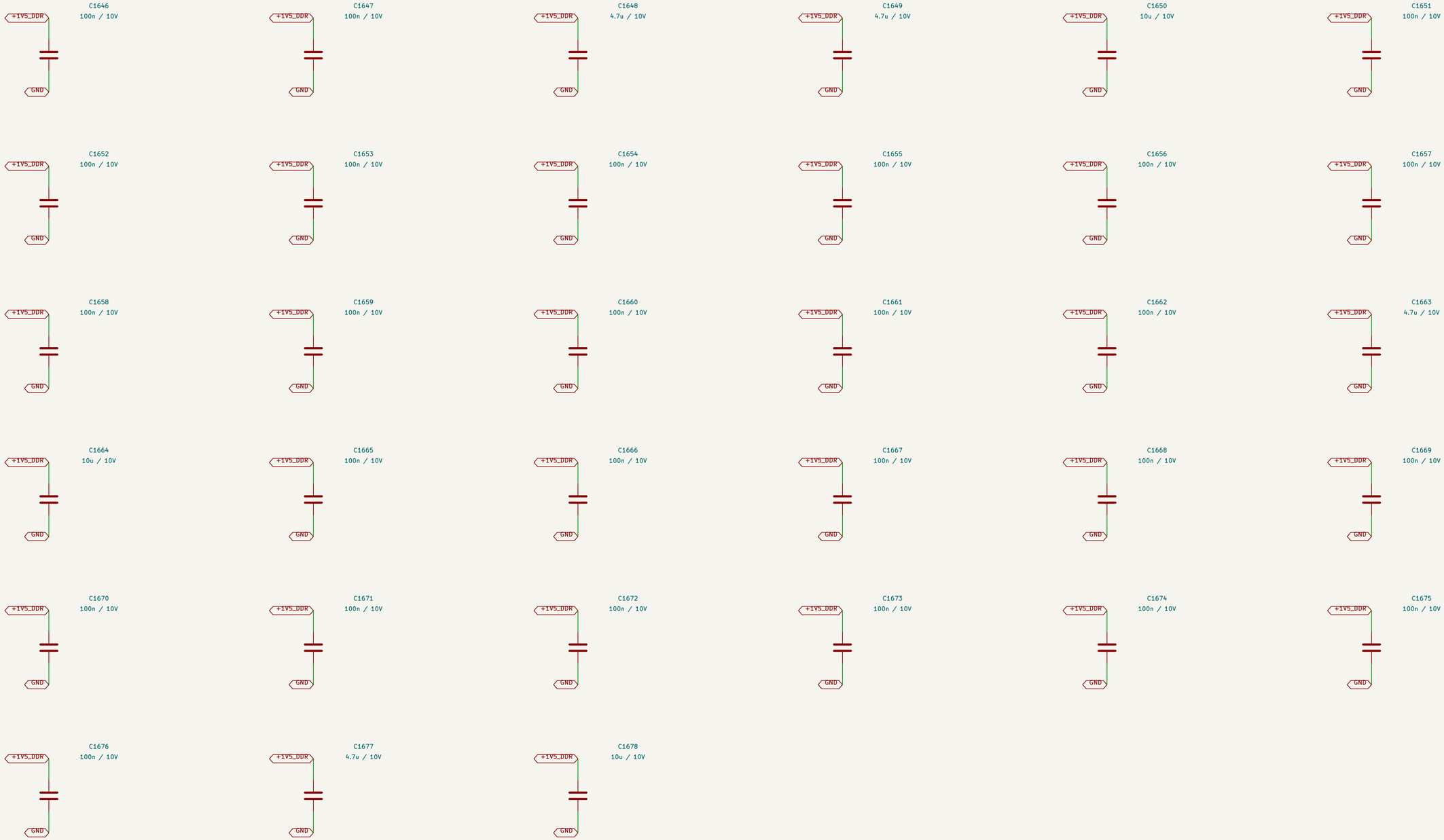
Sources: RK860 Rev1.2 pp3-7; RK3568 AIOT reference p23; Hardware Design Guide V1.2 pp41,60.

Not for fabrication

Sheet: /cpu-power/
File: cpu-power.kicad_sch

**Title: Phoenix G1 - CPU DVFS supply**

| | | |
|---|---|---|
| Size: A3 | Date: | **Rev: DRAFT** |
| KiCad E.D.A. 10.0.7 | | Id: 4/31 |




DRAFT: exact MLCC MPNs, effective capacitance, ESR and mounting inductance not yet qualified.

CPU: six local 22uF plus four regulator 22uF = 220uF nominal, before tolerance/DC bias/temperature.

DDR: first group at SoC; next groups at U2 and U3. Layout must distribute bypass by power-ball groups.

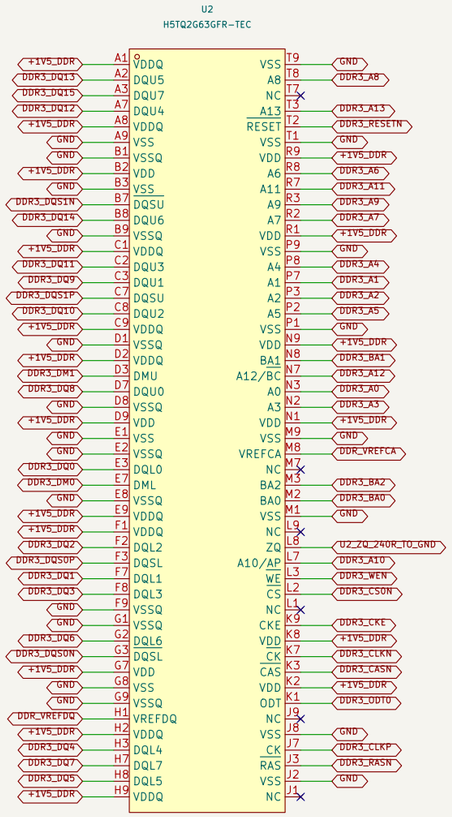
Core values follow Rockchip reference; DDR per-chip allocation is provisional and requires PI review.

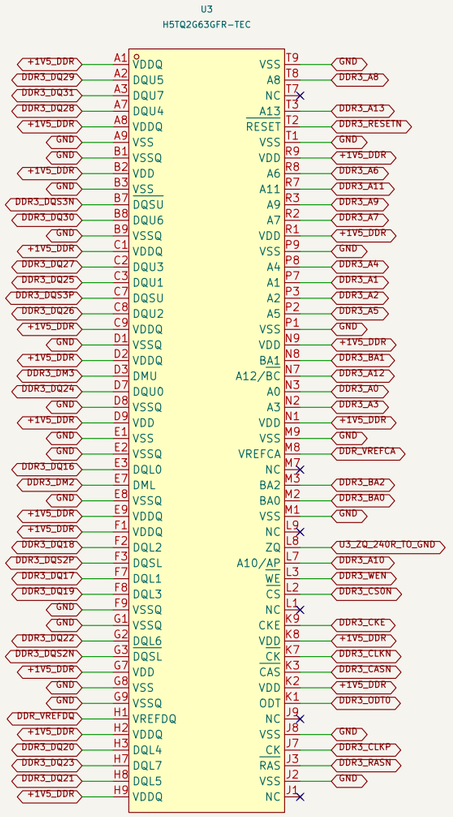
Not for fabrication

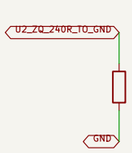
Sheet: /ddr-bypass/
File: ddr-bypass.kicad_sch

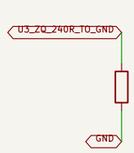
**Title: Phoenix G1 – ddr-bypass**

| | | |
|---|---|---|
| Size: A3 | Date: | **Rev: DRAFT** |
| KiCad E.D.A. 10.0.7 | | Id: 5/31 |
















DDR3 DRAFT: 2 x H5TQ2G63GFR-TEC, 512 MiB, 32-bit bus, one rank. NO FABRICATION RELEASE.

All 192 RAM ball dispositions checked against manufacturer ballout and RK3568 DDR3 alternate functions.

DDR_VREFDQ is driven by RK3568 ball P8; do not connect it to the VREFCA divider.

Both VDD and VDDQ require 1.50 V. This H5TQ part is not a 1.35 V DDR3L device.

RZQ 120R 1% from RK3568 H7 to GND belongs on the SoC sheet; not implemented here.

Power decoupling, clock termination and topology review remain mandatory before routing.

VREF filtering placement and distributed decoupling must follow final physical topology.

Initial target 533 MHz CK / 1066 MT/s; no training or throughput measurement performed.

Passive values are design requirements; exact MPNs and availability have not been qualified.

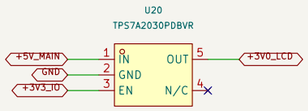
Sources: Hynix Rev1.2 p6; RK3568 PinOut V1.1; Rockchip Hardware Design Guide V1.2.

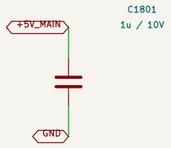
Not for fabrication

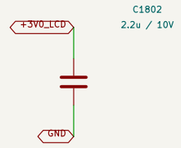
Sheet: /ddr3-memory/
File: ddr3-memory.kicad_sch

**Title: Phoenix G1 – 512 MiB DDR3 memory**

| Size: A3 | Date: | **Rev: DRAFT** |
|---|---|---|
| KiCad E.D.A. 10.0.7 | | Id: 6/31 |

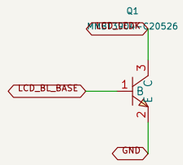


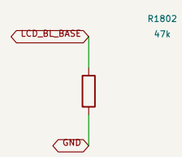



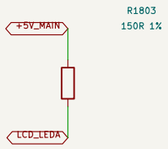



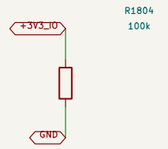







Display 3.0V supply enabled only with IO power; 1.5% TI accuracy fits the stated 2.5–3.3V operating range.

LED drive is current-limited by 150R; brightness varies with VBUS and Vf. PWM dimming; off at reset.

At 5.5V input, Vf=2.9V and zero VCE, current <=17.51mA with 1% R; below 20mA.

Nominal LED current is about 12mA; datasheet full-brightness minimum 15mA is NOT guaranteed.

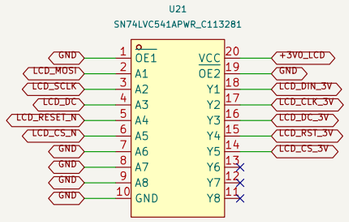
R1804 discharges the enable rail weakly. Validate all supply fall times and no back-powering.

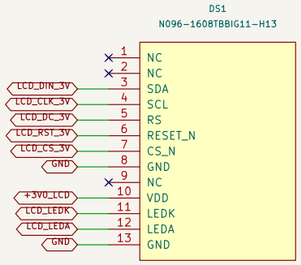
Not for fabrication

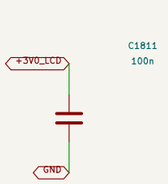
Sheet: /display-power/
File: display-power.kicad_sch

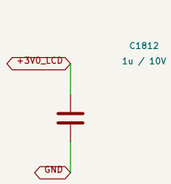
**Title: Phoenix G1 – display supply and backlight**

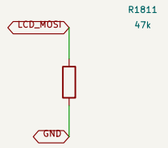
| Size: A3 | Date: | **Rev: DRAFT** |
|---|---|---|
| KiCad E.D.A. 10.0.7 | | Id: 7/31 |

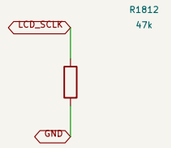


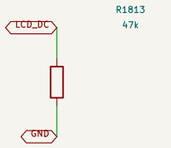



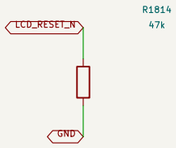



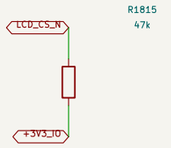




SN74LVC541A accepts 3.3V input while powered at 3.0V. Unused inputs grounded; unused outputs NC.

Display SDA is write-only in this circuit; no readback through the unidirectional buffer.

RELEASE HOLD: display datasheet names VDDIO but has no separate pin. Confirm internal supply relationship.

RELEASE HOLD: 13 contact FPC at 0.7mm pitch is not a standard 0.5mm ZIF. Attachment/footprint remain unqualified.

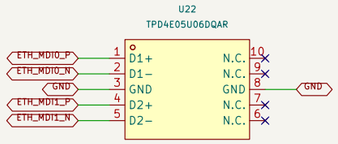
Panel body 13.5 x27.95mm plus flex. Fold radius, exposed contacts, retention and assembly method unresolved.

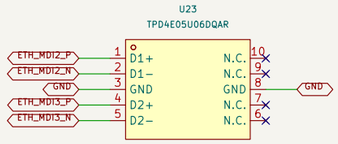
Not for fabrication

Sheet: /display-signals/
File: display-signals.kicad_sch

**Title: Phoenix G1 – write-only SPI display interface**

| Size: A3 | Date: | **Rev: DRAFT** |
|---|---|---|
| KiCad E.D.A. 10.0.7 | | Id: 8/31 |




Two TI C138714 devices protect the eight PHY−side MDI conductors; not the isolated cable side.

Place between transformer and PHY with short ground returns and no long shunt stubs.

Nominal line capacitance 0.5pF; include land pattern and clamp loading in MDI layout review.

5.5V stand−off is not a guarantee that the PHY survives every surge. Check residual stress experimentally.

Cable−side isolation is retained. Do not move these ground−referenced clamps across the transformer.

System ESD/EFT/surge performance requires finished layout and prototype tests; component rating is insufficient.

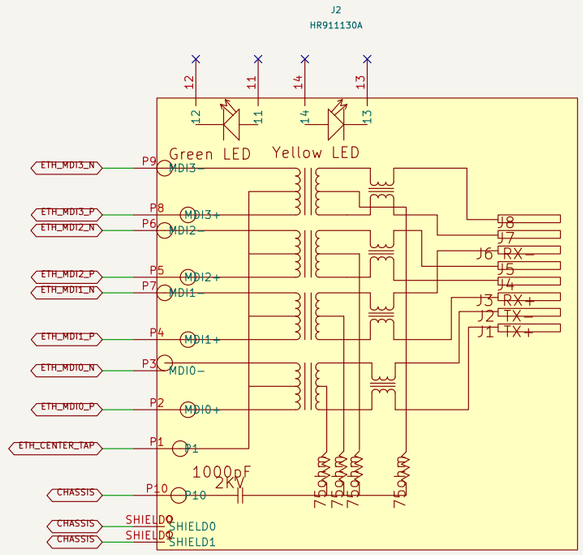
Sources: TI SLVSBO7L pin table, DQA package drawing; TI system−level ESD guide SSZB130 Ethernet topology.

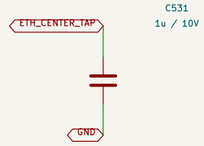
Not for fabrication

Sheet: /ethernet−esd/
File: ethernet−esd.kicad_sch

**Title: Phoenix G1 − PHY−side Ethernet ESD**

| Size: A3 | Date: | **Rev: DRAFT** |
|---|---|---|
| KiCad E.D.A. 10.0.7 | | Id: 9/31 |




DRAFT MAGNETICS. HanRun HR911130A has four chip−side center taps internally connected to P1.

P1 is capacitively grounded; never connect this voltage−mode PHY center tap to 3.3V.

Qualification hold: confirm shared−center−tap configuration for the exact RTL8211F silicon/BSP.

HR911130C Rev B ALSO has a common P1: it is not a separate−center−tap replacement.

P10 carries the integrated cable−side Bob Smith capacitor return; terminate at chassis.

LED pins intentionally unused. Cable−side safety capacitor and isolation are inside the jack.

PHY−side ESD clamps are on ethernet−esd. Chassis bonding and system surge qualification remain open.

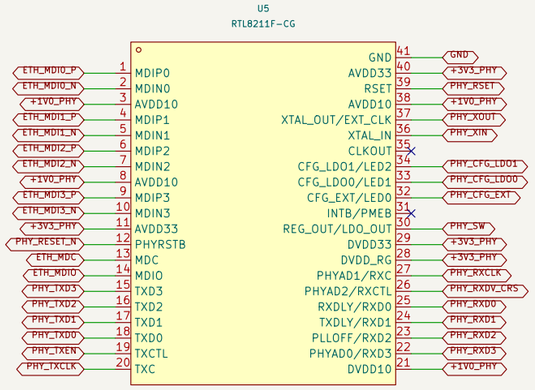
No fabrication release until exact magnetics compatibility, footprint and protection review pass.

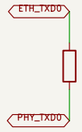
Not for fabrication

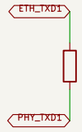
Sheet: /ethernet−magnetics/
File: ethernet−magnetics.kicad_sch

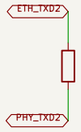
**Title: Phoenix G1 − RJ45 magnetics interface**

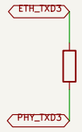
| | | |
|---|---|---|
| Size: A3 | Date: | **Rev: DRAFT** |
| KiCad E.D.A. 10.0.7 | | Id: 10/31 |

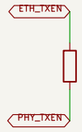


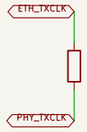



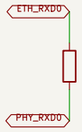



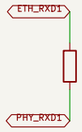



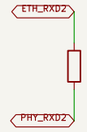



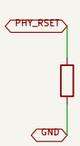



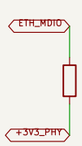






















RGMII at 3.3V. Place TX series resistors at SoC, RX series resistors at PHY.

PHY reset held low by GPIO reset state and R516; firmware must hold >=10ms then deassert.

PHY inserts 2ns TX and RX delay: configure phy-mode=rgmii-id; avoid adding duplicate MAC delay.

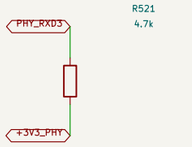
Timing, pad geometry, return paths and power sequencing remain to be qualified.

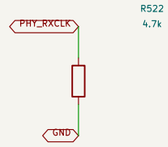
Not for fabrication

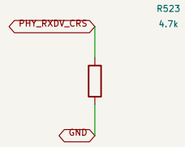
Sheet: /ethernet-phy/
File: ethernet-phy.kicad_sch

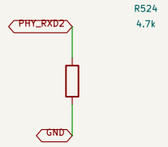
**Title: Phoenix G1 – Gigabit Ethernet PHY**

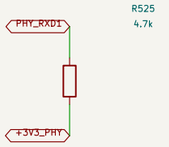
| Size: A3 | Date: | Rev: DRAFT |
|---|---|---|
| KiCad E.D.A. 10.0.7 | | Id: 11/31 |

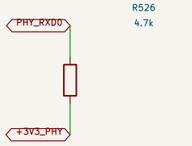


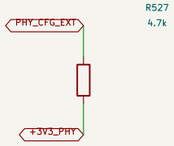



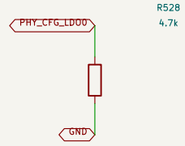



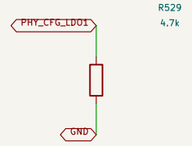



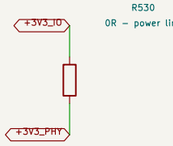



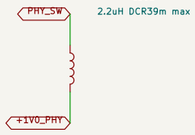



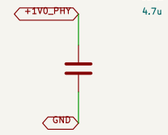



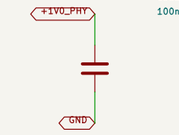



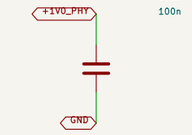



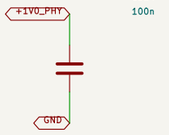



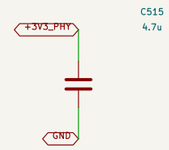



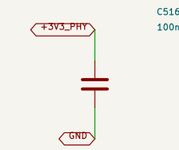



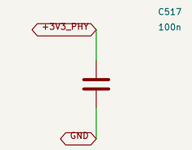



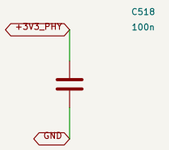



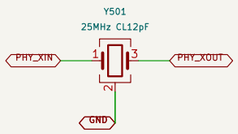



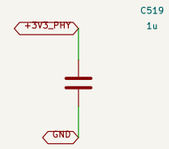



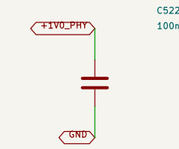



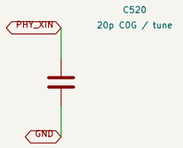



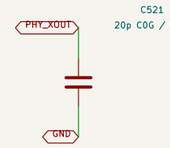




Straps: address 1, external 3.3V RGMII, TX/RX delay ON, PLL-off mode OFF.

C9006 crystal CL12pF: 20pF assumes 2pF stray. Verify drive and frequency on prototype.

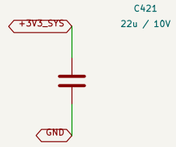
Each AVDD/DVDD pin needs local bypass; verify start-up and discharge against Realtek app note.

Not for fabrication

Sheet: /ethernet-support/
File: ethernet-support.kicad_sch

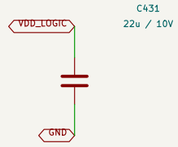
**Title: Phoenix G1 – PHY supplies, straps and clock**

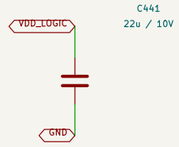
| Size: A3 | Date: | **Rev: DRAFT** |
|---|---|---|
| KiCad E.D.A. 10.0.7 | | Id: 12/31 |





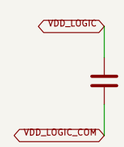


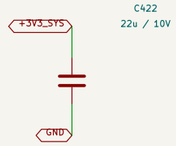




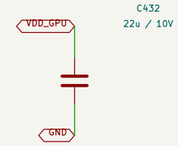


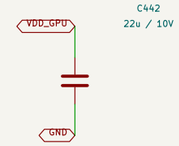




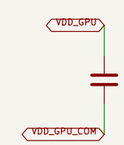


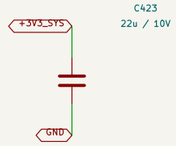




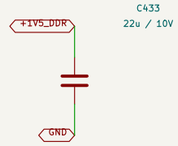


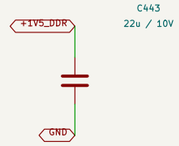




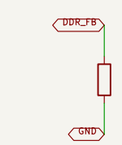


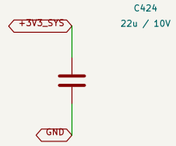




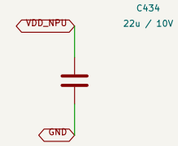


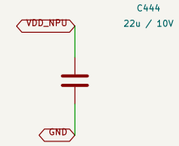




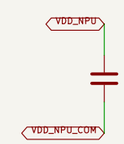


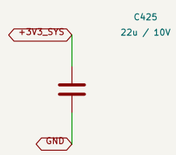




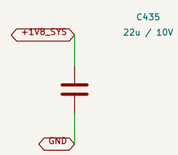


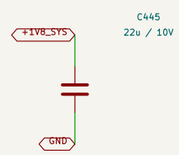









DRAFT. Each local input capacitor belongs beside the corresponding PMIC VCC/SW pair.

DDR feedback targets 1.500V. Static min/max 1.4686/1.5314V; thermal drift and transients not included.

Output capacitors are nominal values: qualify effective capacitance under DC bias before release.

BUCK4/NPU defaults off; GPU retained powered for initial bring-up. Software must match RK809-5 defaults.

Remote-sense nodes must route to the SoC rails, separate from switching paths.

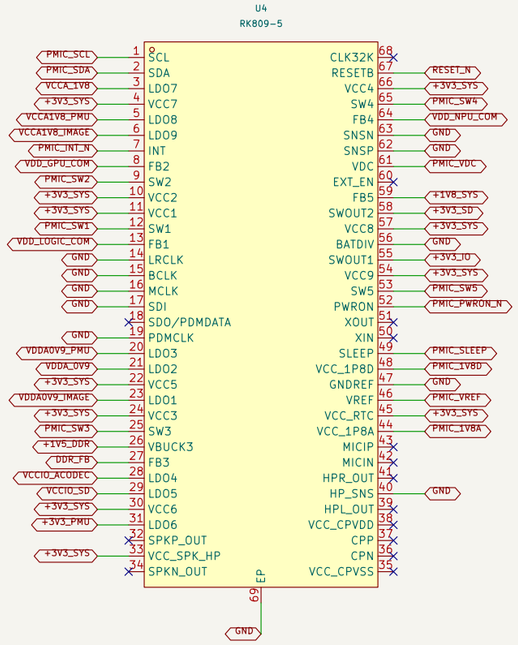
470nH APH201610TR47M C5349593: 85C maximum INCLUDING self-heating. Thermal screening required.

Not for fabrication

Sheet: /pmic-bucks/
File: pmic-bucks.kicad_sch

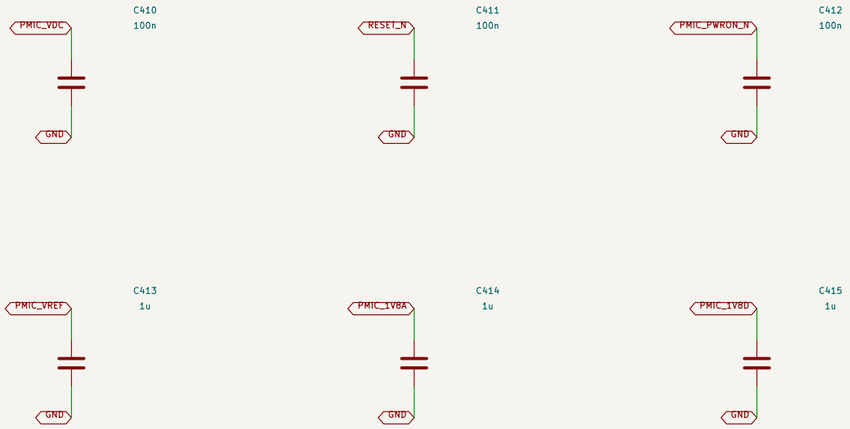
**Title: Phoenix G1 — PMIC switching supplies**

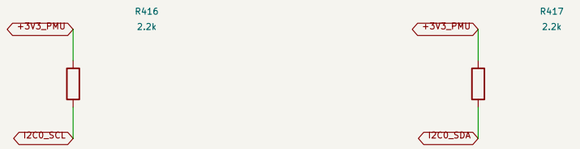
| Size: A3 | Date: | **Rev: DRAFT** |
|---|---|---|
| KiCad E.D.A. 10.0.7 | | Id: 13/31 |










DRAFT: RK809-5 ONLY. VCC_RTC and VCC9 supplied together from +3V3_SYS.

Internal RC oscillator used; no accurate RTC or timed wakeup promised (datasheet 4.4.6).

Audio unused pins grounded/open per RK3568 reference p22. Audio supply retained at 3.3V.

PMIC_SLEEP needs defined SoC reset-state behavior. VDC edge startup must be bench verified.

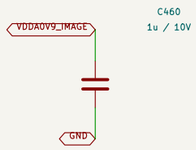
No current-qualified USB input or upstream 3.3V regulator is included on this sheet.

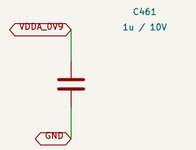
Not for fabrication

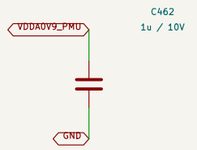
Sheet: /pmic-control/
File: pmic-control.kicad_sch

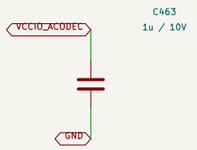
**Title: Phoenix G1 – RK809-5 control and sequencing**

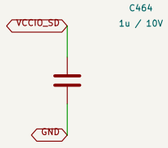
| Size: A3 | Date: | Rev: DRAFT |
|---|---|---|
| KiCad E.D.A. 10.0.7 | | Id: 14/31 |

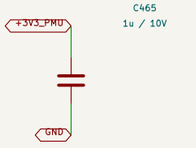


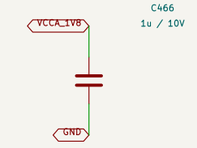



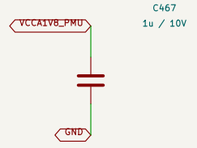



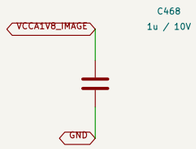



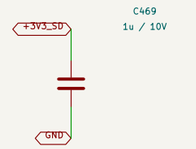



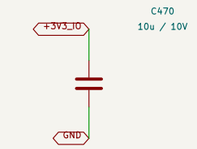



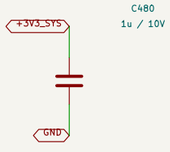



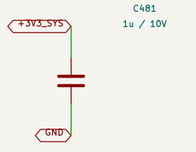



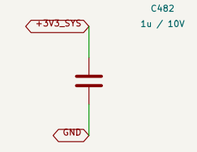



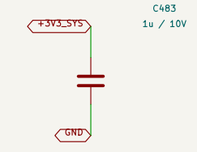



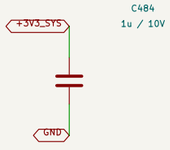



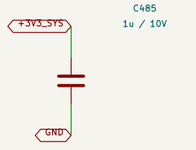




C480..C485 belong locally at VCC5, VCC6, VCC7, VCC8, VCC_RTC, VCC_SPK_HP respectively.

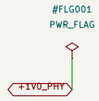
DRAFT: disabled output domains retained with bypass capacitors; full rail/load budget remains open.

Not for fabrication

Sheet: /pmic-ldos/
File: pmic-ldos.kicad_sch

**Title: Phoenix G1 – PMIC linear supplies and bypass**

| Size: A3 | Date: | **Rev: DRAFT** |
|---|---|---|
| KiCad E.D.A. 10.0.7 | | Id: 15/31 |




U5.30 via L501




U4.55 via R530




U17.1 via L1301




U10.D3 via L1001




U10.D3 via R1001,L1001



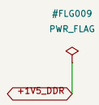

U4.12 via L401



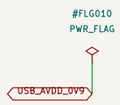

U4.9 via L402



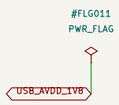

U4.65 via L404



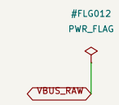

U4.25 via L403




U4.21 via R1601




U4.3 via R1602




J1 USB source via external USB cable




J1 USB return via external USB cable

Flags describe sources through passive R/L paths that KiCad does not propagate. No ERC exclusions are used.

Evidence: review/integrated-connectivity.json. Not a guarantee of source state, sequencing, current or stability.

Not for fabrication

Sheet: /power-source-model/
File: power-source-model.kicad_sch

**Title: Phoenix G1 – explicit ERC power-source model**

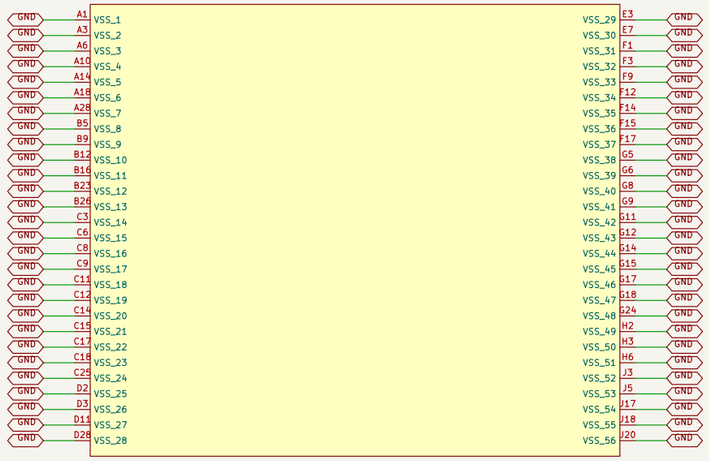
| Size: A3 | Date: | Rev: DRAFT |
|---|---|---|
| KiCad E.D.A. 10.0.7 | | Id: 16/31 |

U1A
RK3568



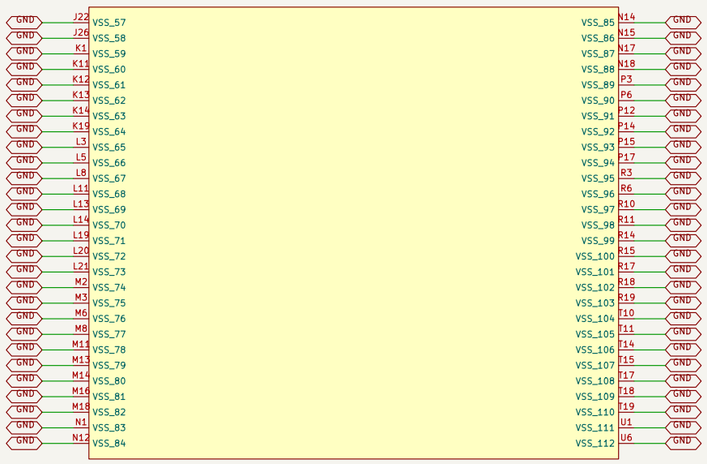

CPU/Logic/GPU/NPU Power

U1B
RK3568




VSS 1

U1C
RK3568




VSS 2

U1D
RK3568




VSS 3

DRAFT. Pin inventory complete; support circuits, decoupling and physical routing remain mandatory.

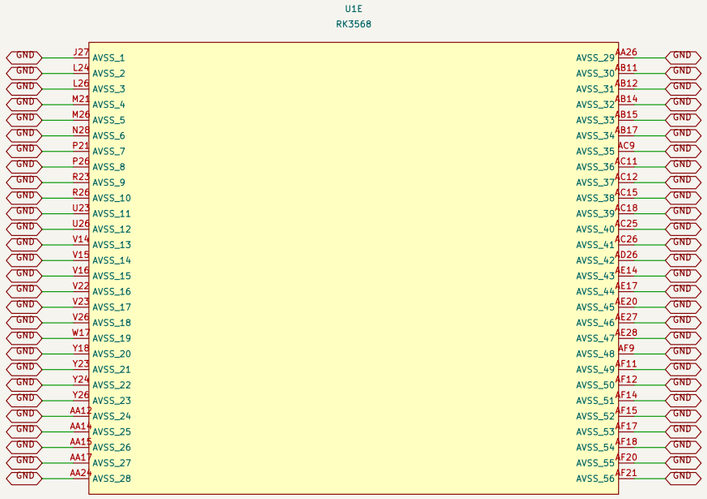
Unused interfaces must be disabled in DTS. STORAGE_* and UNRESOLVED_* are not release-ready nets.

Not for fabrication

Sheet: /soc-01/
File: soc-01.kicad_sch

**Title: Phoenix G1 – RK3568 ball dispositions**

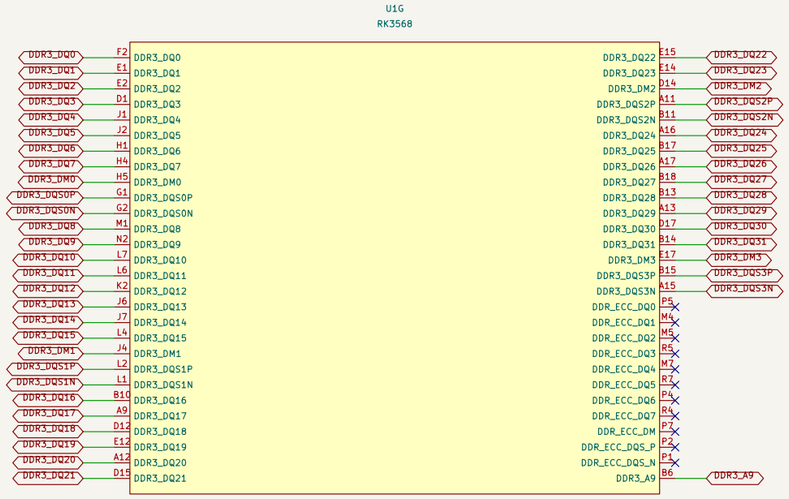
| Size: A3 | Date: | Rev: DRAFT |
|---|---|---|
| KiCad E.D.A. 10.0.7 | | Id: 17/31 |



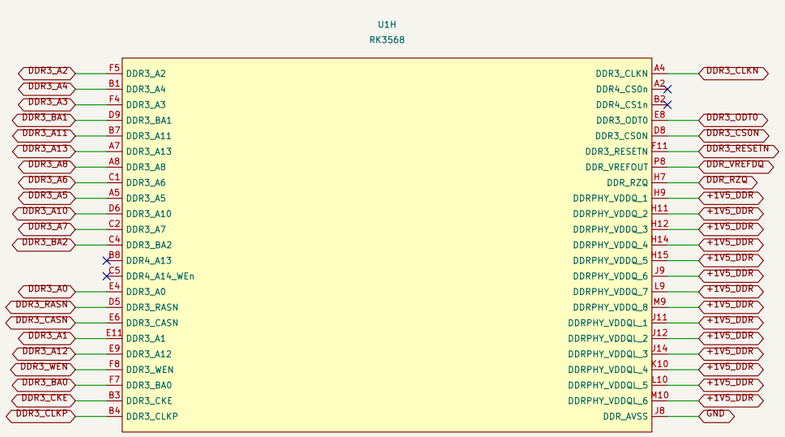

AVSS 1




AVSS 2




DDR 1




DDR 2

DRAFT. Pin inventory complete; support circuits, decoupling and physical routing remain mandatory.

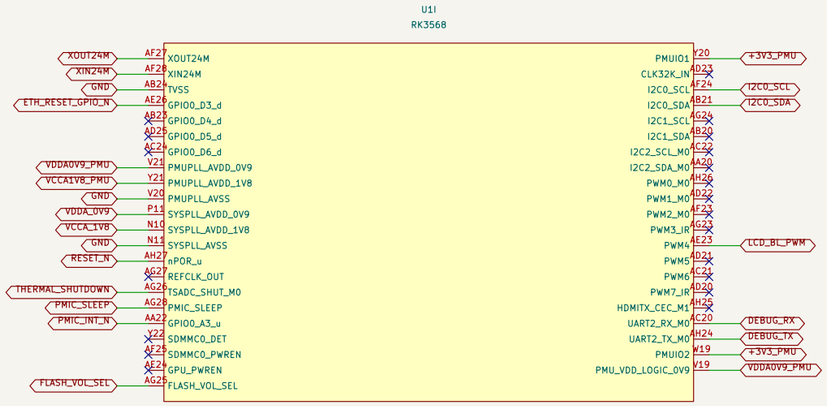
Unused interfaces must be disabled in DTS. STORAGE_* and UNRESOLVED_* are not release-ready nets.

Not for fabrication

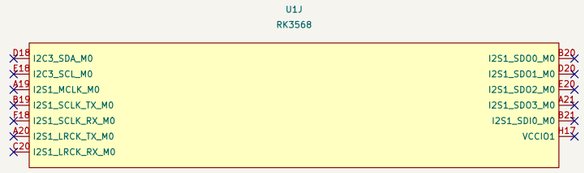
Sheet: /soc-02/
File: soc-02.kicad_sch

**Title: Phoenix G1 — RK3568 ball dispositions**

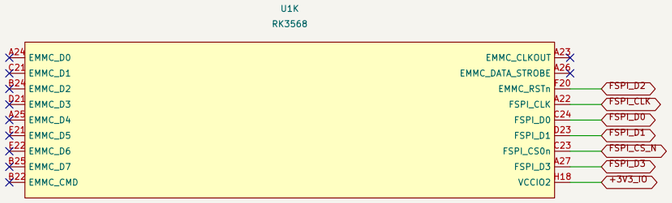
| Size: A3 | Date: | Rev: DRAFT |
|---|---|---|
| KiCad E.D.A. 10.0.7 | | Id: 18/31 |



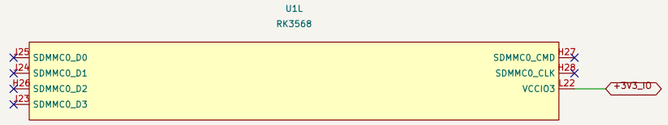

OSC/PLL/PMUIO1/PMUIO2




VCCIO1 Domain




VCCIO2 Domain




VCCIO3 Domain

DRAFT. Pin inventory complete; support circuits, decoupling and physical routing remain mandatory.

Unused interfaces must be disabled in DTS. STORAGE_* and UNRESOLVED_* are not release-ready nets.

Not for fabrication

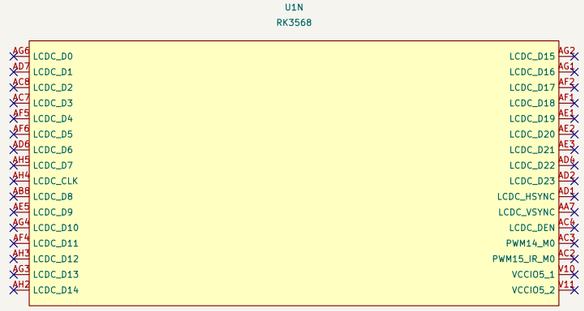
Sheet: /soc-03/
File: soc-03.kicad_sch

**Title: Phoenix G1 – RK3568 ball dispositions**

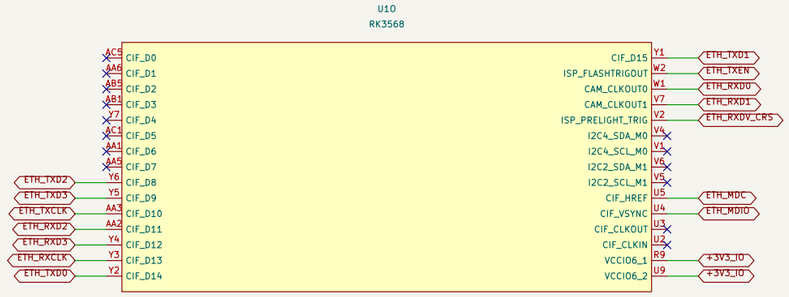
| Size: A3 | Date: | **Rev: DRAFT** |
|---|---|---|
| KiCad E.D.A. 10.0.7 | | Id: 19/31 |



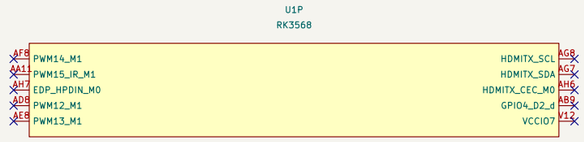

VCCIO4 Domain




VCCIO5 Domain




VCCIO6 Domain




VCCIO7 Domain

DRAFT. Pin inventory complete; support circuits, decoupling and physical routing remain mandatory.

Unused interfaces must be disabled in DTS. STORAGE_* and UNRESOLVED_* are not release-ready nets.

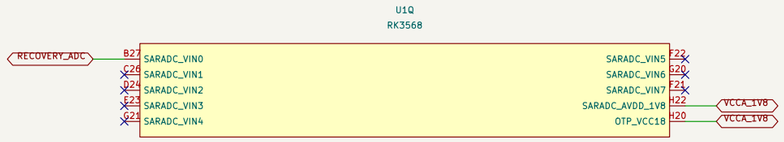
Not for fabrication

Sheet: /soc-04/
File: soc-04.kicad_sch

**Title: Phoenix G1 – RK3568 ball dispositions**

| Size: A3 | Date: | **Rev: DRAFT** |
|---|---|---|
| KiCad E.D.A. 10.0.7 | | Id: 20/31 |

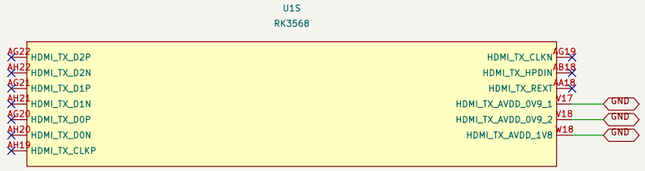
## SARADC/OTP



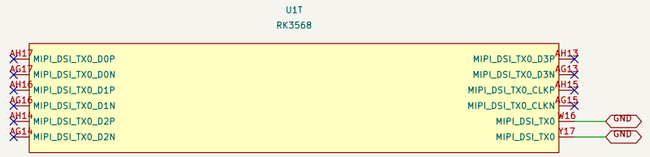

## MIPI_CSI_RX




## HDMI2.0 TX




## MIPI_DSI_TX0/LVDS_TX0




DRAFT. Pin inventory complete; support circuits, decoupling and physical routing remain mandatory.

Unused interfaces must be disabled in DTS. STORAGE_* and UNRESOLVED_* are not release-ready nets.

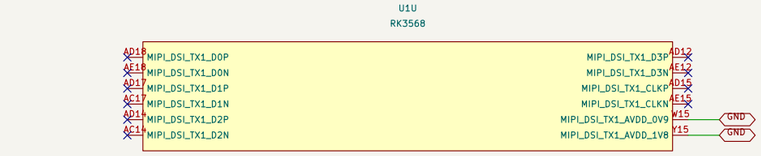
Not for fabrication

Sheet: /soc-05/
File: soc-05.kicad_sch

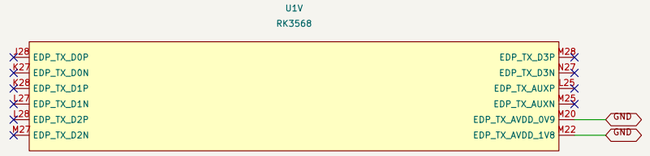
**Title: Phoenix G1 – RK3568 ball dispositions**

| Size: A3 | Date: | **Rev: DRAFT** |
|---|---|---|
| KiCad E.D.A. 10.0.7 | | Id: 21/31 |

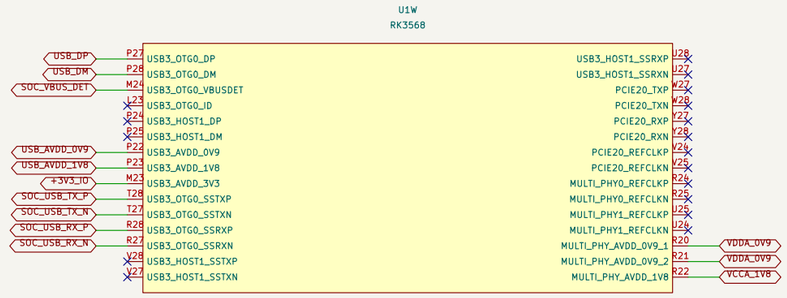
## MIPI_DSI_TX1



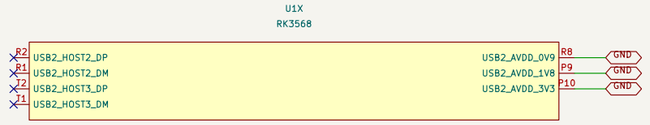

## EDP TX




## USB3.0 OTG0/USB3.0 HOST1/SATA/QSGMII/PCIe2.0




## USB2.0 HOST2/USB2.0 HOST3




DRAFT. Pin inventory complete; support circuits, decoupling and physical routing remain mandatory.

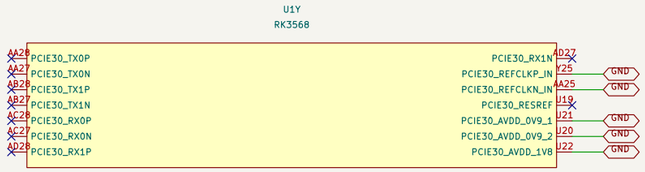
Unused interfaces must be disabled in DTS. STORAGE_* and UNRESOLVED_* are not release-ready nets.

Not for fabrication

Sheet: /soc-06/
File: soc-06.kicad_sch

**Title: Phoenix G1 — RK3568 ball dispositions**

| Size: A3 | Date: | **Rev: DRAFT** |
|---|---|---|
| KiCad E.D.A. 10.0.7 | | Id: 22/31 |




PCIe3.0

DRAFT. Pin inventory complete; support circuits, decoupling and physical routing remain mandatory.

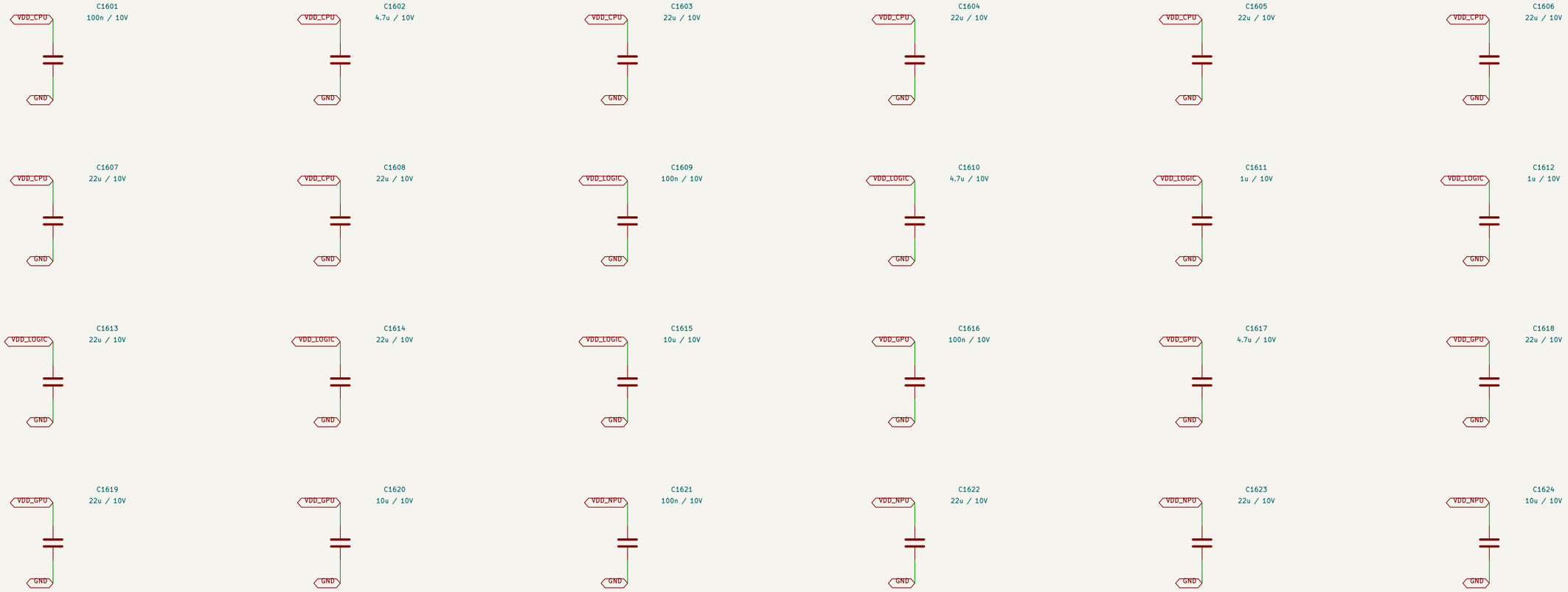
Unused interfaces must be disabled in DTS. STORAGE_* and UNRESOLVED_* are not release-ready nets.

Not for fabrication

Sheet: /soc-07/
File: soc-07.kicad_sch

**Title: Phoenix G1 – RK3568 ball dispositions**

| Size: A3 | Date: | **Rev: DRAFT** |
|---|---|---|
| KiCad E.D.A. 10.0.7 | | Id: 23/31 |




DRAFT: exact MLCC MPNs, effective capacitance, ESR and mounting inductance not yet qualified.

CPU: six local 22uF plus four regulator 22uF = 220uF nominal, before tolerance/DC bias/temperature.

DDR: first group at SoC; next groups at U2 and U3. Layout must distribute bypass by power-ball groups.

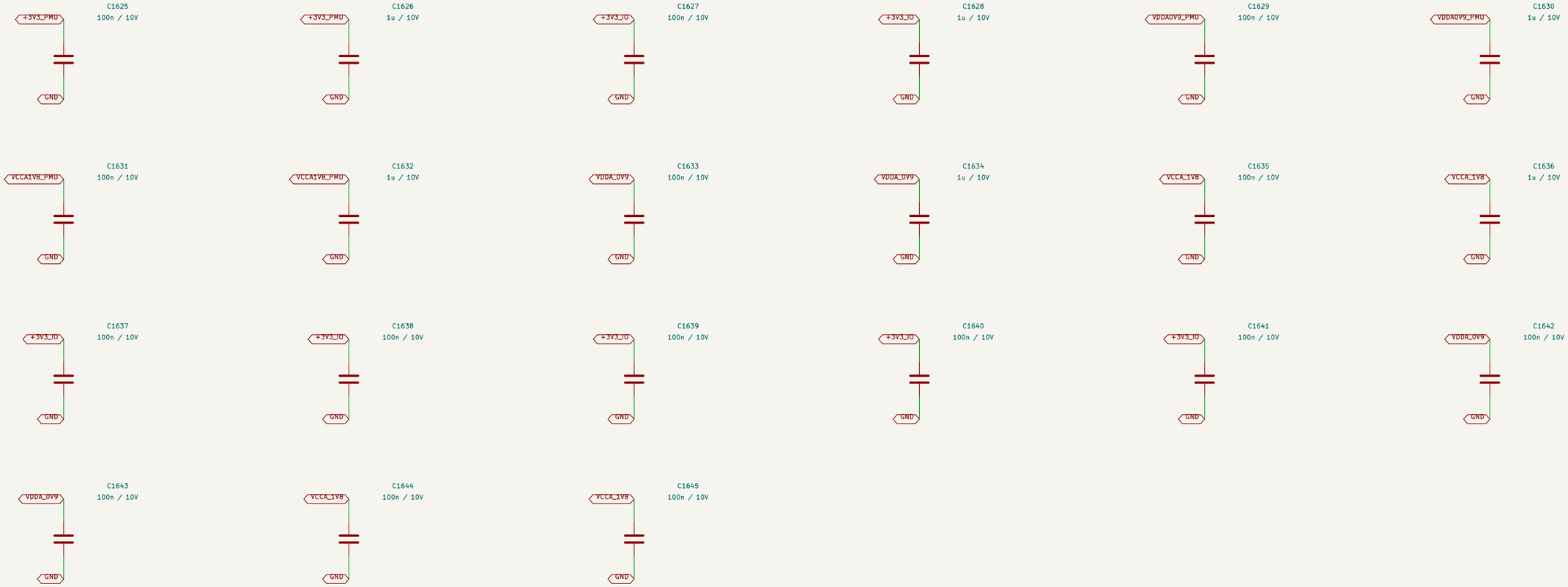
Core values follow Rockchip reference; DDR per-chip allocation is provisional and requires PI review.

Not for fabrication

Sheet: /soc-core-bypass/
File: soc-core-bypass.kicad_sch

**Title: Phoenix G1 - soc-core-bypass**

| | | |
|---|---|---|
| Size: A3 | Date: | **Rev: DRAFT** |
| KiCad E.D.A. 10.0.7 | | Id: 24/31 |




DRAFT: exact MLCC MPNs, effective capacitance, ESR and mounting inductance not yet qualified.

CPU: six local 22uF plus four regulator 22uF = 220uF nominal, before tolerance/DC bias/temperature.

DDR: first group at SoC; next groups at U2 and U3. Layout must distribute bypass by power-ball groups.

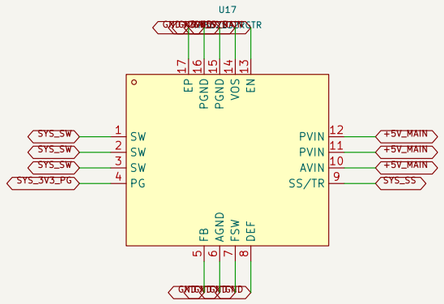
Core values follow Rockchip reference; DDR per-chip allocation is provisional and requires PI review.

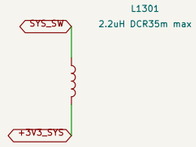
Not for fabrication

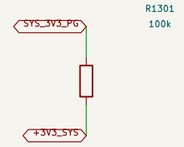
Sheet: /soc-io-bypass/
File: soc-io-bypass.kicad_sch

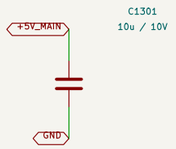
**Title: Phoenix G1 – soc-io-bypass**

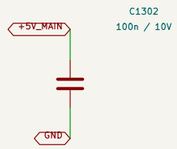
| | | |
|---|---|---|
| Size: A3 | Date: | **Rev: DRAFT** |
| KiCad E.D.A. 10.0.7 | | Id: 25/31 |

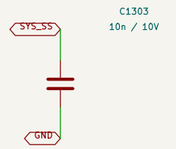


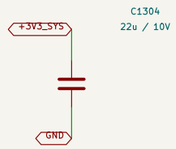



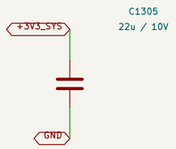




TPS62132 is the fixed 3.3V variant; TPS62133 is 5V and is not interchangeable here.

FSW=GND: 2.5MHz. DEF=GND: nominal output. FB=GND per fixed-voltage recommendation.

SS=10nF gives 5ms typical ramp. Include all PMIC input capacitors in converter stability review.

APH0518T2R2M C5349673: 2.2uH, Isat table 4.98A, Irms table 4.32A; hot behavior and MLCC bias unqualified.

PG is diagnostic only here. PMIC VCC_RTC and VCC9 must rise together from +3V3_SYS.

A 3A converter rating does not guarantee 3A available from USB after CPU and conversion losses.

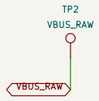
Not for fabrication

Sheet: /system-3v3/
File: system-3v3.kicad_sch

**Title: Phoenix G1 - 3.3V system converter**

| Size: A3 | Date: | **Rev: DRAFT** |
|---|---|---|
| KiCad E.D.A. 10.0.7 | | Id: 26/31 |

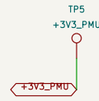






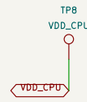










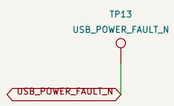






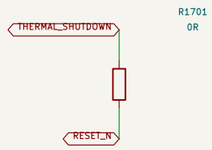











No external header: fixture pads only. UART 3.3V; connect ground first. Do not apply external power to rail pads.

R1701 follows Rockchip hardware TSADC reset. Software must configure active-low shutdown before enabling it.

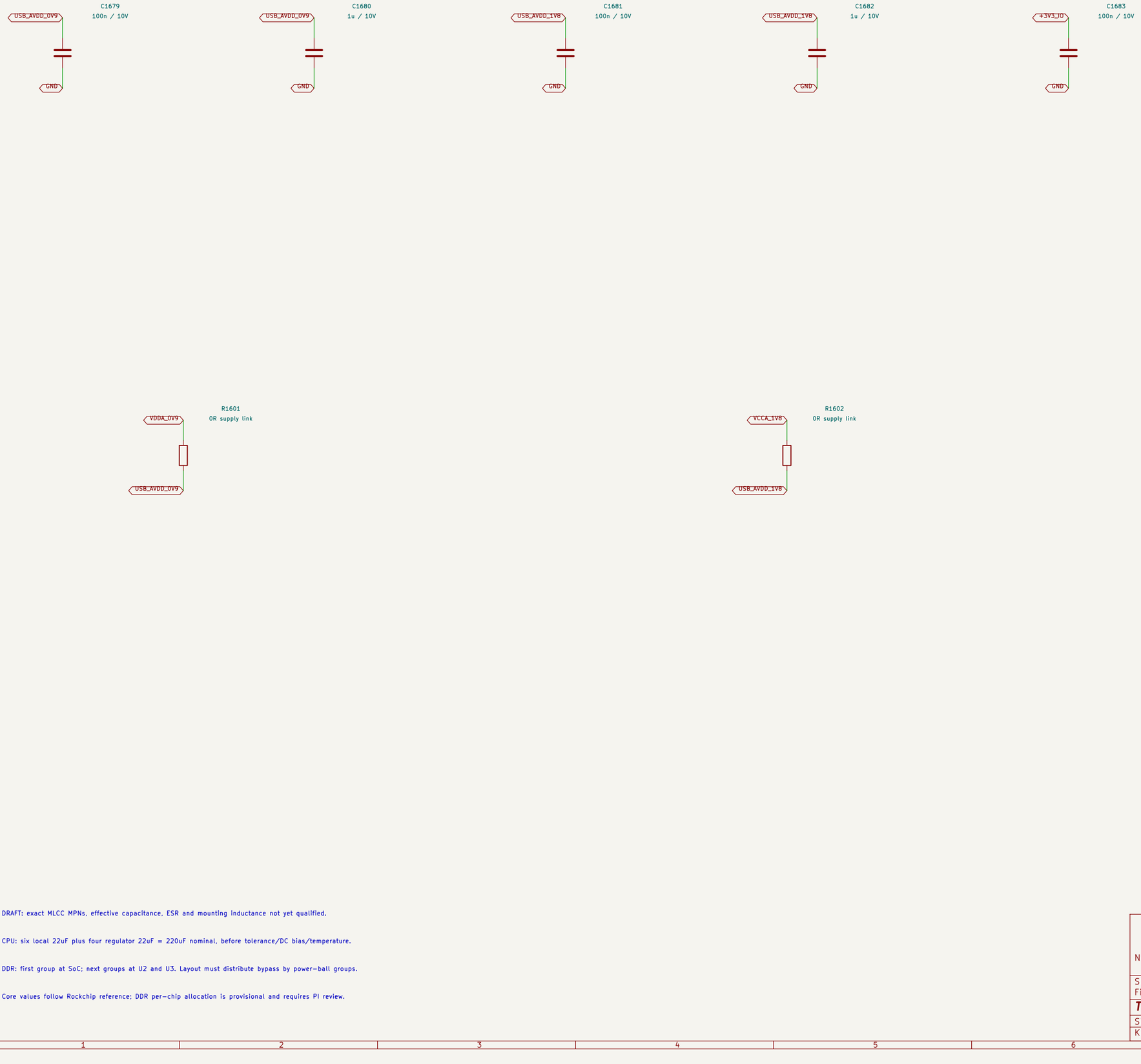
RESET_N also restarts RK809-5. Never add a downstream supervisor that holds it low through the PMIC sequence.

Not for fabrication

Sheet: /test-access/
File: test-access.kicad_sch

**Title: Phoenix G1 — production and recovery pads**

| Size: A3 | Date: | **Rev: DRAFT** |
|---|---|---|
| KiCad E.D.A. 10.0.7 | | Id: 27/31 |




DRAFT: exact MLCC MPNs, effective capacitance, ESR and mounting inductance not yet qualified.

CPU: six local 22uF plus four regulator 22uF = 220uF nominal, before tolerance/DC bias/temperature.

DDR: first group at SoC; next groups at U2 and U3. Layout must distribute bypass by power-ball groups.

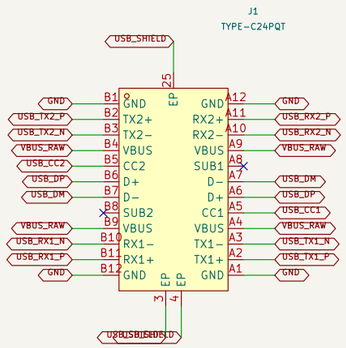
Core values follow Rockchip reference: DDR per-chip allocation is provisional and requires PI review.

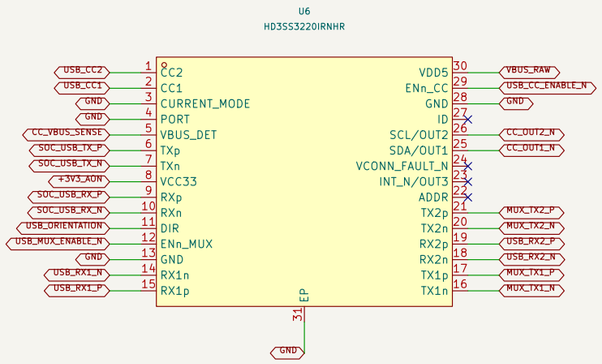
Not for fabrication

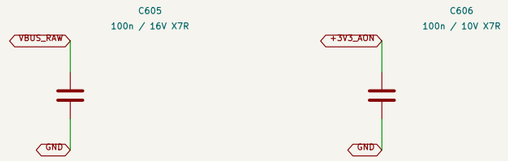
Sheet: /usb-analog-bypass/
File: usb-analog-bypass.kicad_sch

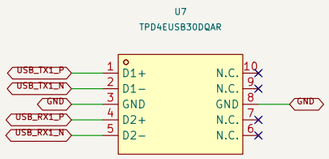
**Title: Phoenix G1 - usb-analog-bypass**

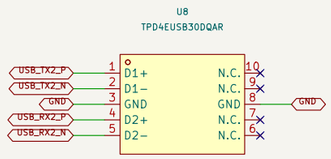
| Size: A3 | Date: | **Rev: DRAFT** |
|---|---|---|
| KiCad E.D.A. 10.0.7 | | Id: 28/31 |

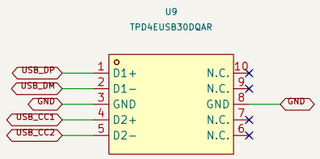


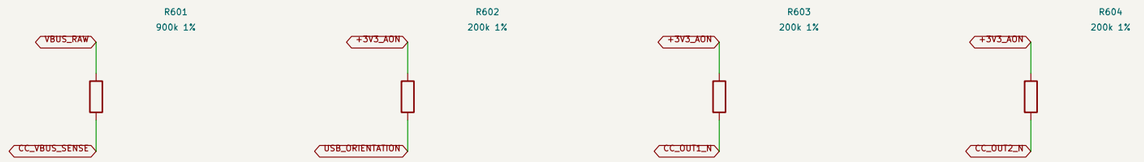










USB-C DATA FRONTEND — ENGINEERING DRAFT, NOT FOR FABRICATION

Sink/UFP only, GPIO mode: PORT=GND, ADDR=NC. No USB Power Delivery negotiation.

Power circuitry is NOT included: VBUS protection, inrush, +3V3_AON and load gating remain required.

Require CC_OUT1_N=0 AND CC_OUT2_N=0 for the proposed 3 A startup policy; OUT2 alone is insufficient.

USB_CC_ENABLE_N requires verified VDD5/VCC33 sequencing (TI section 6.3.12, >=2 ms when controlled).

USB_MUX_ENABLE_N must default disabled until main SoC supplies are valid.

TX AC coupling is on the connector side of the mux, following TI UFP reference. No duplicate SoC TX capacitors.

USB_SHIELD termination and connector footprint remain to validate. Passive MPNs are not selected.

RK3568 endpoints: T28/T27 TX+/−; R28/R27 RX+/−; P27/P28 DP/DM; M24 VBUSDET needs its own sensing circuit.

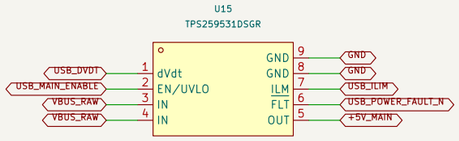
Sources: TI HD3SS3220 SLLSES1E, TPD4EUSB30 SLVSAC2G; Rockchip Hardware Design Guide V1.2.

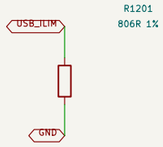
Incomplete system — not for manufacture

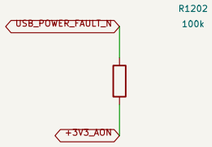
Sheet: /usb-c-frontend/
File: usb-c-frontend.kicad_sch

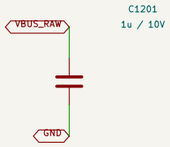
**Title: Phoenix G1 — USB-C data frontend**

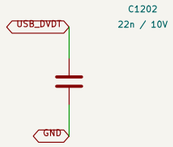
| Size: A3 | Date: | **Rev: DRAFT** |
|---|---|---|
| KiCad E.D.A. 10.0.7 | | Id: 29/31 |

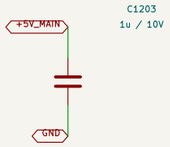



TPS259531: active-high enable, auto-retry, 5.7V typical overvoltage clamp; not an accurate 5V regulator.

RILM=806R targets 2000/R +0.04 = 2.521A typical. Validate guaranteed upper limit with resistor tolerance.

22nF dVdt: nominal 1.91V/ms, approximately 2.62ms to 5V. Check loaded startup/SOA.

Fault output is exposed for diagnostics; auto-retry under overload is not a validated operating mode.

External VBUS transient protection, shutdown discharge and complete power/load budget remain open.

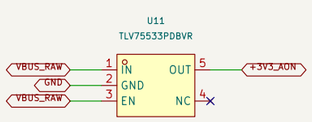
Source: TI TPS2595 SLVSE57C equations 3/4 and electrical-characteristics table.

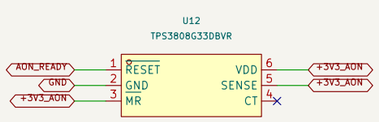
Not for fabrication

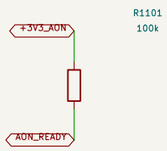
Sheet: /usb-input-protection/
File: usb-input-protection.kicad_sch

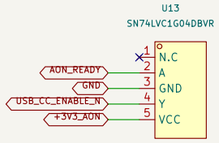
**Title: Phoenix G1 – USB input eFuse**

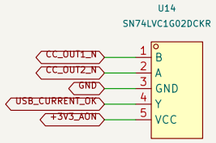
| | | |
|---|---|---|
| Size: A3 | Date: | **Rev: DRAFT** |
| KiCad E.D.A. 10.0.7 | | Id: 30/31 |

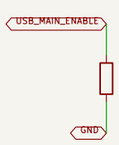


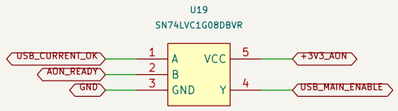



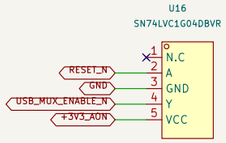



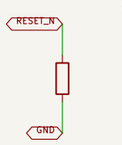



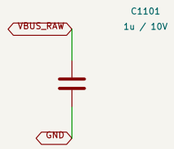



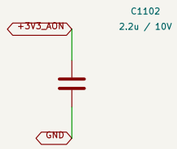



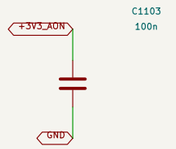



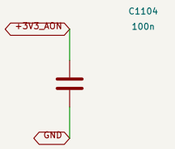



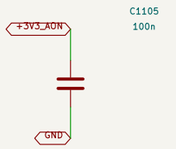



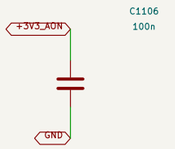



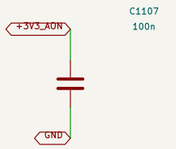




U12 open CT gives 20ms typical delay after 3.3V AON is valid; qualify minimum delay against HD3SS3220.

NOR truth table admits main power only when OUT1=0 AND OUT2=0: Type-C 3A advertisement.

Default-current and 1.5A sources leave main rails off. This design does not negotiate USB PD.

SS mux remains disabled until PMIC releases RESET_N after system rails sequence.

No host-power backfeed path intended. Validate attach/detach, brownout, CC-state changes and USB suspend.

Input +3V3_AON current and total connector-side capacitance require USB compliance verification.

U19 AND gates the current advertisement with AON supervisor readiness; validate ramp behavior.

Not for fabrication

Sheet: /usb-power-control/
File: usb-power-control.kicad_sch

**Title: Phoenix G1 – Type-C current admission and startup**

| Size: A3 | Date: | Rev: DRAFT |
|---|---|---|
| KiCad E.D.A. 10.0.7 | | Id: 31/31 |